

# Harnessing the immunotherapeutic potentials of gamma delta T cells against hematological malignancies

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## Abstract

Gamma delta ( $\gamma\delta$ ) T cells, which constitute about 5%–10% of peripheral blood lymphocytes, play key roles in tumor immunosurveillance and are often enriched within epithelial tissues. They are unique in their Major Histocompatibility Complex-independent antigen recognition via the  $\gamma\delta$  T-cell receptor (TCR) as well as via innate receptors, making them ideal candidates for allogeneic “off-the-shelf” cell therapy products. In humans, two main structural subsets of  $\gamma\delta$  T cells—V $\delta$ 1 and V $\delta$ 2—have been defined, which differ in TCR $\delta$  chains, effector function, and tissue localizations. V $\delta$ 2 T cells constitute the majority of  $\gamma\delta$  T cells in peripheral blood and can be expanded with aminobisphosphonates such as zoledronic acid. In recent years, the potent antitumor functions of V $\delta$ 1 T cells have also been recognized, and new expansion protocols are being developed. Given the ample preclinical evidence of  $\gamma\delta$  T-cell efficacy against hematological malignancies, several  $\gamma\delta$  T-cell-based cell therapy products are currently in clinical development, and there has been an exponential increase in the number of adoptive  $\gamma\delta$  T-cell therapy clinical trials. This comprehensive review provides an overview of the rationale for  $\gamma\delta$  T-cell therapy, ongoing clinical trials, as well as the challenges and future role of  $\gamma\delta$  T-cell-based immune therapies in hematology.

## BACKGROUND

Gamma delta ( $\gamma\delta$ ) T cells usually constitute approximately 5%–10% of peripheral blood (PB) T cells and are a distinct lineage of T lymphocytes which express T-cell receptors (TCRs) composed of gamma ( $\gamma$ ) and delta ( $\delta$ ) chains.<sup>1</sup> The majority of PB T lymphocytes express the  $\alpha\beta$ TCR which usually recognizes peptides presented in the context of self-Major Histocompatibility Complex (MHC) molecules with co-receptors CD4 or CD8 required for  $\alpha\beta$  T-cell activation. Unlike  $\alpha\beta$  T cells, antigen recognition by  $\gamma\delta$  T cells is usually not MHC-restricted and  $\gamma\delta$  TCRs can recognize a wide range of antigens including small molecules, lipids, and whole proteins without the requirement for co-receptor interaction.<sup>2–4</sup> Some classical MHC-restricted  $\gamma\delta$  TCRs have been identified<sup>5</sup> but these are a small minority. In addition to antigen recognition via the TCR,  $\gamma\delta$  T cells possess innate receptors which recognize antigens associated with cellular stress. Therefore,  $\gamma\delta$  T cells belong to a unique group of unconventional T cells with innate and adaptive immune functions, capable of clonal expansion and rapid activation in response to antigen stimulation without prior sensitization.<sup>6</sup> Although identified 40 years ago,<sup>7,8</sup> the crucial functions of these immune cells in health and disease are only being unraveled.

Based on the structure of the variable domains of the  $\delta$  chains, two major subsets of  $\gamma\delta$  T cells have been defined in humans (summarized in Table 1). V $\delta$ 2 chains almost invariably associate with V $\gamma$ 9 chains and are thus referred to as V $\gamma$ 9V $\delta$ 2 T cells. They are the best characterized  $\gamma\delta$  T-cell subtype and are most predominant in PB. By contrast, V $\delta$ 1 T cells are more frequent in the skin and within epithelial tissues. Other subsets including V $\delta$ 3 T cells, which are enriched in the liver<sup>9</sup> and gut epithelium,<sup>10</sup> are rare. Relatively little is known about V $\delta$ 3 T-cell function but recognition of the Major Histocompatibility Complex Class I-related molecule 1 (MR1) has been reported for the V $\delta$ 3 TCR.<sup>11</sup> The tissue localization of specific subsets suggests a vital role for  $\gamma\delta$  T cells in host defense and immune surveillance. Indeed, mice lacking  $\gamma\delta$  T cells demonstrated impaired clearance of bacterial infections<sup>12–14</sup> due to delayed recruitment of immune cells, impaired mucosal immunoglobulin responses,<sup>15</sup> and rapid development of skin tumors following exposure to carcinogens.<sup>16</sup>

Although the structure of the  $\gamma\delta$ TCR suggests the potential of binding to a broad range of antigens,<sup>17,18</sup> the majority of  $\gamma\delta$ TCR ligands remain elusive. The best characterized ligand of the V $\gamma$ 9V $\delta$ 2 TCR are non-peptide phosphoantigens (PAGs), intermediates of the

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**TABLE 1** Characteristics of the major gamma delta (γδ) T-cell subsets in humans.

| γδ T-cell subset  | Tissue distribution                                                                              | Phenotypic characteristics                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Expansion methods                                                  |
|-------------------|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| Vδ1               | Enriched in skin, gut, and other epithelial tissues.<br>Minor γδ population in peripheral blood. | <ul style="list-style-type: none"> <li>• Memory-like features; clonotypic TCR focusing and rapid clonal expansion in response to antigenic stimulation.</li> <li>• Recognize a wide range of ligands which are mostly undefined. Known ligands include MHC-like molecules including MICA/B, CD1d.</li> <li>• Cytolysis of cancer cell lines.</li> <li>• Expression of NKR2.</li> <li>• Relatively resistant to activation-induced cell death.</li> <li>• Higher expression of PD-1 and TIGIT.</li> </ul> | Anti-CD3, Vδ1 agonist antibodies, and stimulatory cytokines.       |
| Vδ2 (also Vγ9Vδ2) | Predominant in peripheral blood.                                                                 | <ul style="list-style-type: none"> <li>• Semi-invariant TCR pairing with Vγ9 chains.</li> <li>• Recognition of intracellular accumulation of phosphoantigens in "stressed" cells.</li> <li>• Professional antigen-presentation function.</li> <li>• Cytolysis of cancer cell lines.</li> <li>• Expression of NKR2.</li> <li>• Little known; some MR1 reactivity and IL-17 production.</li> </ul>                                                                                                         | Zoledronate and stimulatory cytokines including IL-2 and/or IL-15. |
| Vδ3               | Liver and gut epithelium. Rare in peripheral blood.                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Not defined.                                                       |

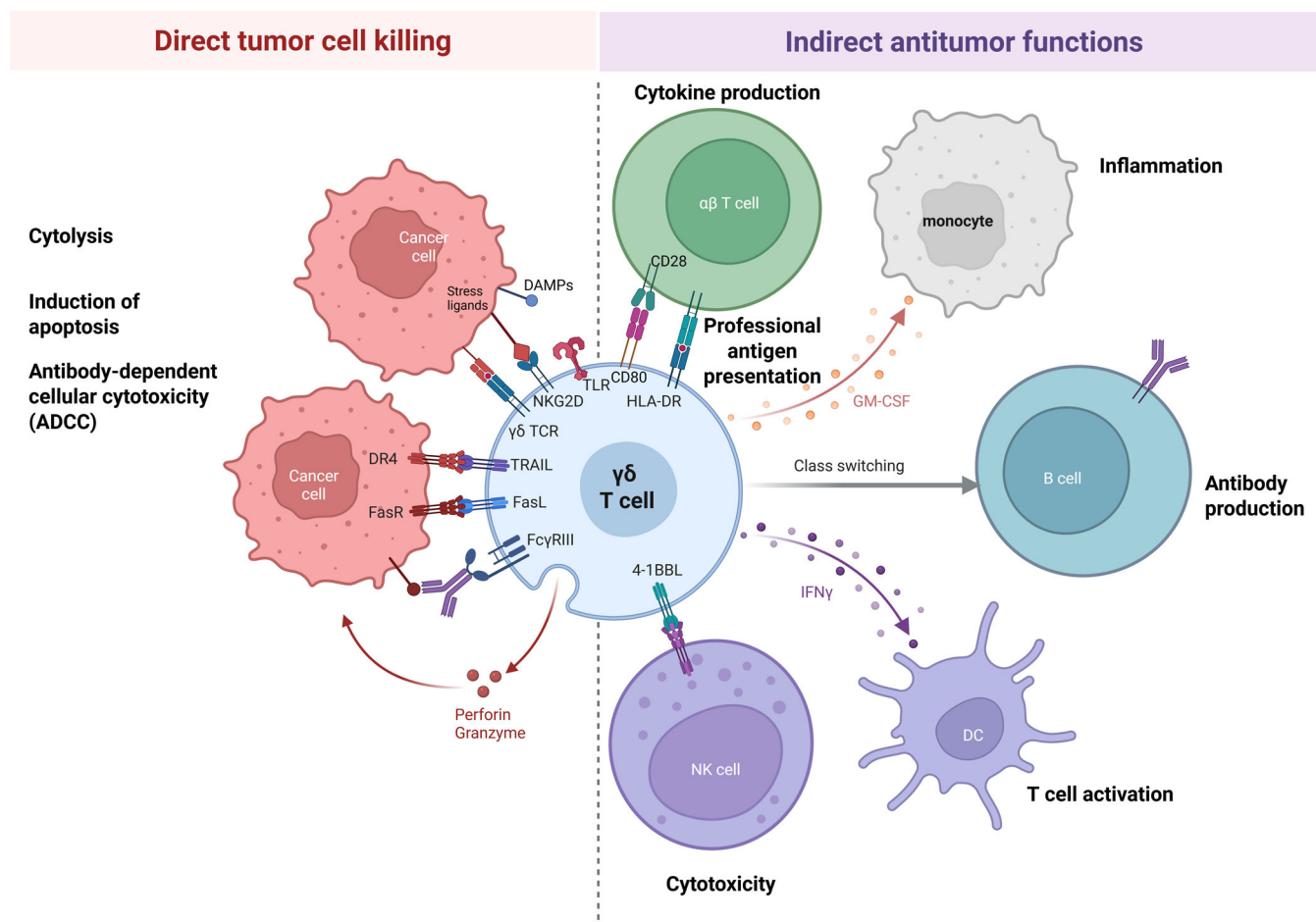
Abbreviations: IL, interleukin; MHC, Major Histocompatibility Complex; MICA/B, MHC Class I chain-related protein A and B; MR1, Major Histocompatibility Complex Class I-related molecule 1; NKR, natural killer-cell receptor; PD-1, programmed cell death 1 receptor; TCR, T-cell receptor; TIGIT, T-cell immunoreceptor with immunoglobulin and ITIM domain.

isoprenoid (mevalonate) biosynthetic pathway.<sup>19</sup> The most potent known pAg activator of the Vγ9Vδ2 TCR is the microbial precursor of isopentenyl pyrophosphate (IPP), (E)-4-hydroxy-3-methyl-but-2-enyl pyrophosphate (HMB-PP), produced by gram-negative bacteria and *Mycobacterium tuberculosis*.<sup>20</sup> In human cells, accumulation of IPP generated via both the classical and alternative mevalonate pathways occurs in states of cellular stress and in transformed cells.<sup>21</sup> The inhibition by aminobisphosphonates of farnesyl pyrophosphate synthase, a key enzyme in the mevalonate pathway which converts IPP to farnesyl pyrophosphate, leads to intracellular accumulation of IPP and is the basis of many expansion methods for Vγ9Vδ2 T cells.<sup>22</sup> The mechanism of Vγ9Vδ2 TCR recognition of intracellular IPP accumulation has been the subject of intense study in recent years. Crucial to this mechanism are the butyrophilins (BTNs), a family of Type I membrane proteins with two immunoglobulin-like extracellular domains and structural homology to the B7 superfamily of proteins.<sup>23</sup> Intracellular PAgS bind to the intracellular B30.2 domain of BTN3A1 inducing a conformational change<sup>24–26</sup> which permits it to associate with BTN2A1.<sup>27</sup> The resulting BTN3A1–BTN2A1 complex then binds to and activates the Vγ9Vδ2 TCR.<sup>27,28</sup> While BTN2A1 binds directly to germline-encoded regions of the Vγ9 chain,<sup>29</sup> both BTN3A1 and BTN2A1 are required for TCR activation and the loss of BTN2A1 cannot be compensated for by BTN3A1.<sup>28</sup> Recognition of the role of BTNs in the activation of Vγ9Vδ2 T cells has led to the development of an activating humanized BTN3A antibody which is currently in early phase clinical trials against a range of cancer types.<sup>30</sup> In contrast to Vδ2 T cells, most ligands of the Vδ1 TCR are unknown but several putative candidates have been identified. These include several MHC superfamily members such as CD1d, which presents lipid molecules<sup>31,32</sup> and MHC Class I chain-related protein A and B (MICA/B).<sup>6</sup> MICA/B are also ligated by natural killer Group 2D (NKG2D) which is expressed by both Vδ1 and Vδ2 T cells. The expression of such innate receptors in addition to TCR specificity make γδ T cells robust immune effectors.

Besides the difference in antigen recognition and tissue localization between Vδ1 and Vδ2 γδ T cells, several phenotypic distinctions have been observed. In contrast to Vδ2, the Vδ1 subset expresses higher levels of C–C chemokine receptor Type 7 (CCR7) and chemokine (CXC) receptor (CXCR)–1, whereas there is higher C–C chemokine receptor Type 5 (CCR5) expression in the former.<sup>33,34</sup> These differences in the expression of homing receptors have been suggested as a possible mechanism for the differential tissue localizations of γδ T-cell subsets. Vδ1 T cells also express higher levels of PD-1 compared to Vδ2 T cells<sup>35</sup>; however, recently published work has shown that PD-1-expressing Vδ1 T cells represent a functional, tissue-adapted state, which is in contrast to the association of PD-1 upregulation with functional exhaustion in cytotoxic αβ T cells.<sup>36</sup> Vδ1 T cells are more resistant than Vδ2 T cells to the immunosuppressive effects of galectin-3<sup>37</sup> and activation-induced cell death.<sup>34</sup> Recognition of the putative tissue adaptations and immunotherapeutic potentials of Vδ1 T cells has triggered recent interests in specific expansion methods for this subset.

## ANTICANCER FUNCTIONS OF γδ T CELLS

γδ T cells mediate anticancer responses via direct and indirect mechanisms (summarized in Figure 1). Upon activation via the γδ TCR and/or other activating receptors including NKG2D, CD94/NKG2C, DNAX accessory molecule-1 (DNAM-1), Nkp30, and Nkp44,<sup>38</sup> γδ T-cells can initiate lysis of target cells by secretion of cytolytic



**FIGURE 1** Schematic illustration of the anticancer properties of gamma delta ( $\gamma\delta$ ) T cells.  $\gamma\delta$  T cells sense a wide range of cellular signals via innate receptors such as the natural killer Group 2D (NKG2D) receptor, which is activated by a range of stress signals and Toll-like receptors, which sense DAMPs, damage-associated molecular patterns. Target cell engagement via the  $\gamma\delta$  TCR is also achieved via a range of ligands that are upregulated in malignant cells and other states of cellular stress. The antitumor activity of  $\gamma\delta$  T cells occurs via both direct mechanisms including the release of cytolytic enzymes such as perforin and granzyme, induction of apoptosis via the Fas ligand (FasL) or tumor necrosis factor-related apoptosis-inducing ligand (TRAIL) as well as via antibody-dependent cellular cytotoxicity. Indirect antitumor mechanisms include activation of other immune cell mediators such as macrophages and dendritic cells (DCs) via secretion of chemokines such as interferon- $\gamma$  (IFN $\gamma$ ), granulocyte-macrophage colony-stimulating factor (GM-CSF).  $\gamma\delta$  T cells also act as professional antigen-presenting cells to  $\alpha\beta$  T cells via the Class II Major Histocompatibility Complex molecule, HLA-DR, and by providing co-stimulation via the CD80-CD28 axis. Class switching of B cells to stimulate antibody production is induced by  $\gamma\delta$  T cells via secretion of IL-4 and IL-10. Activation of NK cells also occurs via the 4-1BB:4-1BB ligand (4-1BBL) interaction. DR4, death receptor 4; FasR, Fas receptor; Fc $\gamma$ RIII, Fc-gamma receptor III. This figure was created with [Biorender.com](#).

proteins such as perforins and granzyme B.<sup>39,40</sup> Experimental evidence from receptor blocking suggests that  $\gamma\delta$  T cells can lyse target cells following ligation of NKG2D, independent of TCR stimulation but lysis of some cancer cell lines requires signaling via the TCR in addition to NKG2D.<sup>41</sup> An alternative pathway for direct cell lysis involves the initiation of apoptosis of target cells via the interaction with FasL and tumor necrosis factor (TNF)-related apoptosis-inducing ligand (TRAIL) with their respective receptors.<sup>42</sup> Furthermore,  $\gamma\delta$  T cells express the low-affinity Fc-gamma receptor CD16 (Fc $\gamma$ RIII), which can initiate antibody-dependent cellular cytotoxicity (ADCC). Experimental evidence of ADCC is supported by the observation that  $\gamma\delta$  T-cell cytotoxicity is enhanced following the addition of rituximab and trastuzumab to cultures with CD20+ and HER2+ target cells, respectively.<sup>43</sup>  $\gamma\delta$  T cells also mediate anticancer responses indirectly by the activation and recruitment of other immune cells via secretion of cytokines and acting as professional antigen presenting cells (APCs).<sup>44</sup> Given their versatility and lack of MHC restriction,  $\gamma\delta$  T

cells are excellent candidates for adoptive cell therapy against cancers particularly as an off-the-shelf product.

## EXPERIMENTAL EVIDENCE OF $\gamma\delta$ T-CELL ACTIVITY IN HEMATOLOGICAL MALIGNANCIES

The antitumor reactivity of  $\gamma\delta$  T cells has been demonstrated in several experimental models of hematological malignancies. Early in vitro experiments demonstrated effector functions for unfractionated  $\gamma\delta$  T cells activated with IL-2, against cancer cell lines including the Daudi Burkitt's lymphoma-derived cell line, which lacks expression of MHC Class I.<sup>45</sup> The observation that the addition of blocking antibodies against HLA A, B, and C did not diminish their cytotoxic activity first gave credence to the MHC-independent mode of antigen recognition by  $\gamma\delta$  T cells.<sup>46</sup> While most early in vitro experiments demonstrating anticancer properties of  $\gamma\delta$  T cells involved

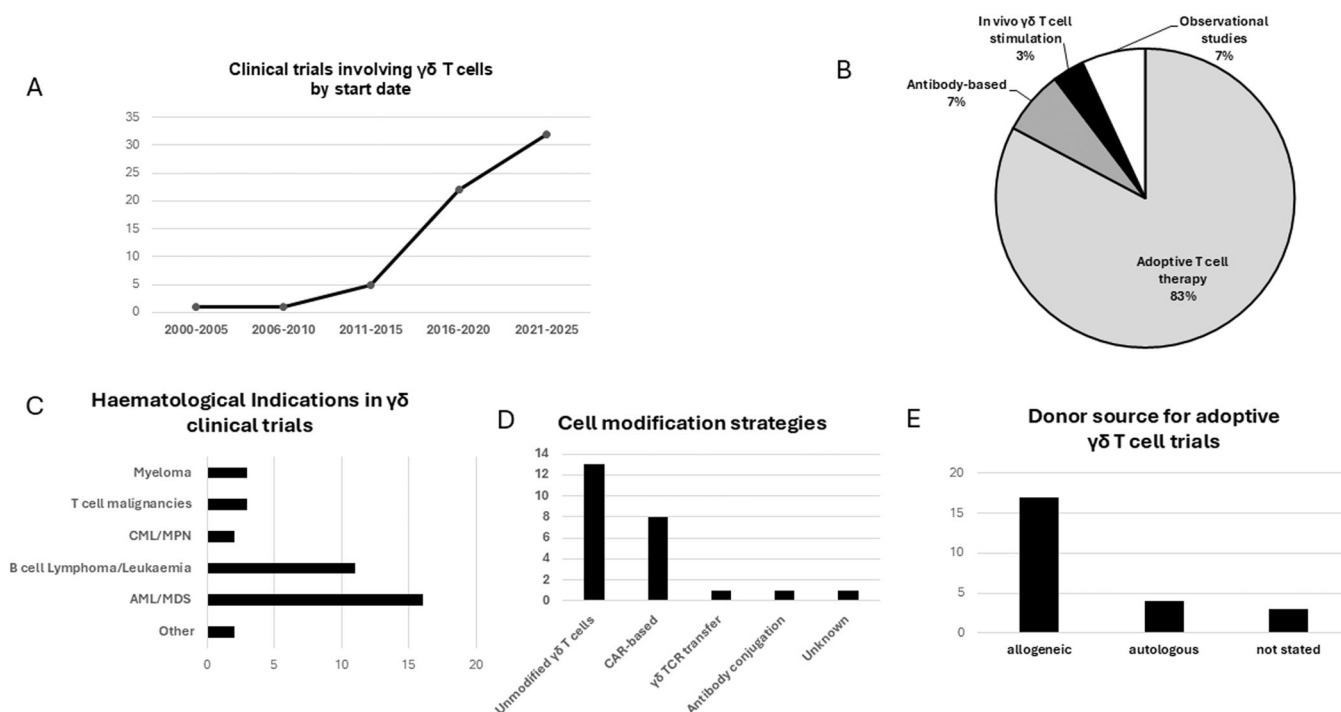
unfractionated (predominantly V $\delta$ 2 subset), the V $\delta$ 1 subset has also been shown to possess potent anticancer properties.  $\gamma\delta$  T cells of the V $\delta$ 1 subset expanded with concanavalin A (Con A) demonstrated specific cytotoxicity against chronic lymphocytic leukemia (CLL) cells in vitro.<sup>47</sup> Similarly, phytohemagglutinin (PHA)-expanded V $\delta$ 1 T cells were cytotoxic against myeloma cell lines and primary bone-marrow-derived malignant plasma cells.<sup>48</sup> More recently, clinically relevant methods based on cytokines and agonist antibodies have been developed for the expansion of the V $\delta$ 1 subset. Purified V $\delta$ 1 T cells expanded with a cocktail of cytokines ("DOT cells") were cytotoxic against CLL cell lines, autologous and allogeneic primary CLL cells but spared autologous healthy leukocytes.<sup>49</sup> A simple expansion method using anti-CD3 (OKT3) and IL-15 resulted in the expansion of V $\delta$ 1 T cells, which were cytotoxic against neuroblastoma and leukemia cell lines in vitro but not reactive against allogenic PBMCs.<sup>50</sup>

One of the earliest clinical indications of the role of  $\gamma\delta$  T cells in immune surveillance was from the observation of an association between improved leukemia free survival (LFS) and increased  $\gamma\delta$  T-cell frequency (>10% of T cells in PB) following partial T cell depleted allogeneic stem cell transplantation (SCT) for acute leukemia.<sup>51</sup> An update of the study, including long-term (8-year) follow-up, confirmed that T-cell recovery with increased  $\gamma\delta$  T cells (as opposed to normal or low levels) was associated with significantly improved 5-year overall survival (OS) (70.8% vs. 19.6%,  $P < 0.0001$ ) and 5-year LFS (54.4% vs. 19.1%,  $P = 0.0003$ ). No increased incidence of acute graft versus host disease (GVHD) was observed in the high  $\gamma\delta$  T-cell frequency group. Similar findings were observed in the pediatric post-SCT setting ( $n = 102$ ).<sup>52</sup> Notably, in addition to improved event-free survival, elevated  $\gamma\delta$  T-cell reconstitution was associated with a significantly decreased cumulative incidence of infections (0.18% vs. 0.53%;  $P = 0.02$ ). An analysis from cohorts of newly diagnosed acute myeloid leukemia (AML) patients identified an association between

low V $\gamma$ 9V $\delta$ 2 T-cell frequency and significantly inferior 5-year overall and relapse-free survivals.<sup>53</sup> Furthermore, Gentles et al. analyzed genomic signatures from about 18,000 tumors and 39 malignancies, identifying tumor-infiltrating  $\gamma\delta$  T cells as a strong positive predictor of survival.<sup>54</sup> These findings provide strong evidence for the role of  $\gamma\delta$  T cells in tumor immunosurveillance. Furthermore, their MHC-independent mechanism of antigen recognition permits graft versus leukemia (GvL) effects without the undesirable GVHD response. Based on the lack of alloreactivity of  $\gamma\delta$  T cells and prior observations of the reduced risk of disease recurrence post-allogeneic SCT, the adoptive transfer of ex vivo expanded donor-derived  $\gamma\delta$  T cells has become an attractive strategy that is being investigated in clinical trials.

## CLINICAL TRIALS OF $\gamma\delta$ T-CELL-BASED CELL THERAPY IN HEMATOLOGICAL MALIGNANCIES

Early attempts at in vivo stimulation of V $\gamma$ 9V $\delta$ 2 T cells with the administration of aminobisphosphonates and IL-2 to patients achieved variable expansions, limited clinical responses, and toxicities arising from systemic cytokine effects.<sup>55,56</sup> More recently, there has been a surge in the number of registered clinical trials of adoptively infused  $\gamma\delta$  T cells (Figure 2). Although the majority of current  $\gamma\delta$  T-cell therapy trials are for solid malignancies, this review will focus on clinical studies for hematological indications. There were 29 ongoing, recently completed, or terminated clinical trials involving  $\gamma\delta$  T cells for hematological conditions listed on [clinicaltrials.gov](https://clinicaltrials.gov) (accessed January 31, 2025), and these are summarized in Table 2. The majority of  $\gamma\delta$  T-cell-related clinical studies were adoptive cell therapy strategies, and the most frequent hematological indication was AML, followed by B-cell malignancies (Figure 2).



**FIGURE 2** Summary of gamma delta ( $\gamma\delta$ ) T cells based clinical trials. (A) Number of  $\gamma\delta$  T-cell-related clinical trials by start date. (B) Proportions of study types for hematology-related  $\gamma\delta$  T-cell trials. (C) Hematological indications in  $\gamma\delta$  T-cell-related clinical trials. (D) Cell modification strategies of  $\gamma\delta$  T-cell-related clinical trials in hematology. (E) Donor source for adoptive  $\gamma\delta$  T-cell clinical trials in hematology. Clinical trial details were accessed via [clinicaltrials.gov](https://clinicaltrials.gov) as of January 31, 2025. AML, acute myeloid leukemia; CML, chronic myeloid leukemia; MDS, myelodysplastic syndrome; MPN, myeloproliferative neoplasm.

## Clinical studies involving in vivo $\gamma\delta$ T-cell stimulation

Several ongoing clinical trials employ strategies for activating  $\gamma\delta$  T cells in vivo including the administration of zoledronic acid (ZOL), cytokines, and/or activating antibodies. The feasibility of in vivo generation and stimulation of  $\gamma\delta$  T cells was investigated in the Phase 1 HILDEGAZ trial.<sup>57</sup> Following a haploidentical allogeneic stem cell transplant for hematological malignancies, 16 trial participants received ZOL and increasing doses of IL-2. A significantly higher median number of V $\gamma$ 9V $\delta$ 2 T-cells was observed during IL-2 treatment compared with non-randomized transplant recipients who had not received ZOL/IL-2. No Grade 3 adverse events (AEs) other than acute GVHD (which resolved after steroid treatment) were observed in the trial cohort. An alternative approach to stimulating  $\gamma\delta$  T cells in vivo involves the administration of a humanized BTN3A antibody (ICT01), which stimulates V $\gamma$ 9V $\delta$ 2 T cells.<sup>30</sup> The ongoing EVICTION Trial of ICT01 includes a cohort of patients with hematological malignancies, but the results of this group have yet to be reported. Interim results of ICT01 in combination with pembrolizumab in a cohort of patients with checkpoint inhibitor-refractory melanoma suggested rapid trafficking of activated V $\gamma$ 9V $\delta$ 2 T-cells and transient, dose-dependent increases in serum cytokines, and an overall favorable safety profile.<sup>58,59</sup> Furthermore, a novel single-domain bispecific antibody (LAVA-051),<sup>60</sup> which engages tumor-associated CD1d with the V $\delta$ 2 TCR was evaluated in a Phase I trial (NCT04887259). Preliminary results<sup>61,62</sup> of the trial involving multiple myeloma (MM) and CLL patients indicated safety and tolerability of LAVA-051 at the evaluated doses; however, the trial has been terminated by the sponsors for commercial reasons.

## Adoptive $\gamma\delta$ T-cell therapy trials

The safety and feasibility of allogeneic  $\gamma\delta$  T cells expanded from healthy donors was investigated in Phase 1 TCB-202-001 (NCT03790072) Trial.<sup>63</sup> Following a standard lymphodepleting conditioning regimen, trial participants with AML ineligible for allogeneic SCT received V $\gamma$ 9V $\delta$ 2  $\gamma\delta$  T cells, which had been expanded from haploidentical donors with ZOL and IL-2. Three dose levels of  $\gamma\delta$  T cells,  $10^6$ /kg,  $10^7$ /kg, and  $10^8$ /kg, were evaluated with no GVHD or treatment-related AEs of Grade 3 or greater. Although the trial was terminated early due to the COVID-19 pandemic, response assessments suggested some preliminary evidence for efficacy in AML. One of the four patients who had a bone marrow assessment on Day 28 post-cell infusion achieved a complete response (CR). Other trials involving the administration of allogeneic  $\gamma\delta$  T cells have confirmed the safety of this approach with no GVHD observed in a prior pilot study<sup>64</sup> and in recent studies.<sup>64,65</sup>

Chimeric antigen receptor (CAR)-transduced and other modified  $\gamma\delta$  T cells are currently in early-phase clinical trials. ADI-001, an allogeneic V $\delta$ 1 CAR T-cell therapy product targeting CD20, is being evaluated against B-cell malignancies (NCT04735471, GLEAN-1 study). Early results from the Trial which comprised 50% of patients who had progressed on previous CD19-directed CAR T-cell therapies, indicated ORR and CR rates of 71% and 63%, respectively,<sup>66</sup> and robust in vivo expansion of ADI-001 with levels significantly higher in trial subjects achieving a best overall response of PR or better.<sup>67</sup> There was no association between the degree of patient-donor HLA mismatch and response, cell expansion, or persistence by Day 28. Given the safety profile and early signal of efficacy of ADI-001, it is understood that the trial sponsors are considering a pivotal Phase 2 Trial. CAR-modified  $\gamma\delta$  T-cell products targeting CD19, CD7, and NKG2DL<sup>68</sup> are also being investigated in early-phase trials. An alternative T-cell-redirection strategy relevant to  $\gamma\delta$  T-cell antitumor reactivity is the use of TCR transfer, an approach originally developed by Gada Therapeutics. This strategy is being investigated in a trial of

$\gamma\delta$ TCR transduced T cells (NCT04688853) in patients with relapsed/refractory myeloma. In the Phase I study, trial participants receive autologous (mainly  $\alpha\beta$ ) T cells transduced with a high-affinity V $\gamma$ 9V $\delta$ 2 TCR with broad antitumor reactivity.<sup>69</sup>

The role of adoptive transfer of  $\gamma\delta$  T cells to prevent AML recurrence post-allogeneic SCT is being explored in several clinical trials with strategies including transfer of  $\gamma\delta$  T- and natural killer (NK)-cell-enriched ( $\alpha\beta$  T-cell depleted) donor lymphocyte infusion (DLI; NCT05686538, IDEAL trial) and infusion of allograft donor-derived ex vivo activated  $\gamma\delta$  T cells (NCT03533816 and NCT05015426). The administration of Expanded/Activated Gamma-Delta (EAGD, also termed INB-100) T-cell infusions following haploidentical hematopoietic SCT for hematological malignancies is being investigated in a Phase I trial (NCT03533816) with early reports indicating favorable safety and tolerability profiles.<sup>70</sup> Recent updated results released by IN8bio indicated remarkable clinical responses in the AML cohort with 100% of trial participants remaining in complete remission after a median follow-up of 20.1 months with the observation of  $\gamma\delta$  T-cell persistence beyond 12 months.<sup>71</sup> In another Phase I trial of adoptive  $\gamma\delta$  T cells (NCT05015426), the published interim safety report indicated no GVHD, CRS, or immune effector cell-associated neurotoxicity syndrome (ICANS) within the 42-day dose-limiting toxicity period at the two dose levels assessed ( $6.25 \times 10^6$  cells/kg and  $3.13 \times 10^7$  cells/kg). The  $\gamma\delta$  T cells (>90% V $\delta$ 2) infused were obtained from the same stem cell donors, expanded ex vivo with ZOL and IL-2, and further cultured with artificial antigen-presenting cells. All trial participants were in CR (minimal residual disease negative) at the time of reporting, and updated results are awaited.

Currently, the majority of  $\gamma\delta$  T-cell-based adoptive therapy studies are early-phase clinical trials, which are limited by small and often nonrepresentative patient populations. Although these trials provide proof of safety and tolerability for  $\gamma\delta$  T-cell-based approaches, large Phase III trials investigating useful efficacy endpoints such as long-term persistence, response rates, and disease-free survival are required to contextualize the therapeutic potential of  $\gamma\delta$  T cells relative to current conventional T-cell-based approaches.

## $\gamma\delta$ T CELLS IN IMMUNOTHERAPY: CONTROVERSIES AND CHALLENGES

Studies have identified V $\delta$ 1  $\gamma\delta$  T cells as the predominant subset in breast,<sup>35</sup> colorectal,<sup>72</sup> gastric,<sup>73</sup> and melanoma<sup>74</sup> tumor biopsies. Although most experimental models demonstrate potent antitumor properties and positive correlation with survival, one study identified a subset of  $\gamma\delta$  T cells (predominantly of the V $\delta$ 1 phenotype producing IL-17,  $\gamma\delta$ T17) which was associated with tumor progression in colorectal cancer by recruiting myeloid-derived suppressor cells.<sup>72</sup> Although the mechanism of polarization of  $\gamma\delta$  T cells towards the IL-17-producing phenotype within human tissues remains unclear, a distinct subset of murine  $\gamma\delta$  T cells that expressed the IL-23 receptor responded to IL-1 $\beta$  and IL-23 stimulation leading to the production of IL-17, independent of TCR stimulation.<sup>75</sup> The IL-23-driven  $\gamma\delta$ T17 differentiation of peripheral  $\gamma\delta$  T cells was also confirmed in another murine model.<sup>76</sup> Furthermore, upon activation, V $\delta$ 1 T cells from adjacent healthy colorectal tissues produced IFN $\gamma$  but hardly any IL-17 or TNF $\alpha$  in contrast to V $\delta$ 1 cells within colorectal cancer biopsies.<sup>77</sup> In another study, increased intratumoral frequencies of IL-17-producing V $\delta$ 1 and V $\delta$ 2 T cells were associated with advanced skin squamous cell carcinoma (SCC) and risk of mortality in contrast to IFN $\gamma$ -producing intratumoral V $\delta$ 1 and V $\delta$ 2 T cells, which were more frequent in SCC patients at earlier disease stages.<sup>78</sup> Recently, using a combination of RNA sequencing and flow cytometry methods,



**TABLE 2** Gamma delta T-cell-based clinical trials in hematology. (continued on next page)

| NCT number                     | Study title                                                                                                                                                                                         | Acronym  | Other IDs                            | Study status                   | Conditions                                  | Interventions                                                                                  |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------------------|--------------------------------|---------------------------------------------|------------------------------------------------------------------------------------------------|
| <i>Adoptive cell therapies</i> |                                                                                                                                                                                                     |          |                                      |                                |                                             |                                                                                                |
| NCT05886491                    | A study of GDX012 in adults with relapsed or refractory acute myeloid leukemia                                                                                                                      |          | TAK-012-1501                         | RECRUITING                     | AML                                         | BIOLOGICAL: GDX012                                                                             |
| NCT05686538                    | Innate donor effector allogeneic lymphocyte infusion after stem cell transplantation: the IDEAL trial                                                                                               | IDEAL    | Protocol VIP14.0.11                  | RECRUITING                     | AML/MDS                                     | OTHER: innate donor lymphocyte infusion (iDLI)                                                 |
| NCT05653271                    | ACE1831 in adult subjects with relapsed/refractory CD20-expressing B-cell malignancies                                                                                                              |          | ACE1831-001                          | RECRUITING                     | B-cell lymphoma; NHL, DLBCL, PMBCL, MZL, FL | BIOLOGICAL: ACE1831 DRUG: obinutuzumab                                                         |
| NCT05554939                    | Allogeneic CD19-targeting CAR-γδ T cell therapy in r/r NHL                                                                                                                                          |          | CHN-PLAGH-BT-072                     | RECRUITING                     | NHL                                         | BIOLOGICAL: allogeneic CD19-CAR-γδ T cell                                                      |
| NCT05388305                    | Universal CAR-γδ T cell injection in the AML patients                                                                                                                                               |          | CAR-γδ T for AML                     | UNKNOWN                        | AML                                         | BIOLOGICAL: CAR-- γδT cells                                                                    |
| NCT05358808                    | ACHIEVE - efficacy and effectiveness of adoptive cellular therapy with ex-vivo expanded allogeneic γδ T-lymphocytes (TCB008) for patients with refractory or relapsed acute myeloid leukaemia (AML) | ACHIEVE  | TCB008-001                           | RECRUITING                     | AML, MDS                                    | BIOLOGICAL: TCB008                                                                             |
| NCT06463327                    | ACHIEVE2 - safety and preliminary efficacy of intravenous TCB008 in patients with relapse or refractory acute myeloid leukemia (AML) or myelodysplastic syndrome (MDS)/AML                          | ACHIEVE2 | TCB008-003                           | NOT YET RECRUITING             | AML, MDS                                    | BIOLOGICAL: TCB008                                                                             |
| NCT05302037                    | Allogeneic NKG2DL-targeting CAR-γδ T cells (CTM-N2D) in advanced cancers (ANGELICA)                                                                                                                 | ANGELICA | CTM-N2D-102                          | RECRUITING                     | Refractory/relapsed cancer                  | BIOLOGICAL: allogeneic NKG2DL-targeting chimeric antigen receptor-grafted γδ T cells (CTM-N2D) |
| NCT05015426                    | Gamma delta T-cell Infusion for AML at high risk of relapse after Allo HCT                                                                                                                          |          | MCC-20305 9BC08                      | ACTIVE NOT RECRUITING          | AML                                         | BIOLOGICAL: gamma delta T-cell infusion                                                        |
| NCT05001451                    | Study of GDX012 in patients with MRD positive AML                                                                                                                                                   |          | GDX012U-001                          | TERMINATED (BUSINESS DECISION) | AML                                         | BIOLOGICAL: GDX012 suspension for IV Infusion                                                  |
| NCT04887259                    | Trial of LAVA-051 in patients with relapsed/refractory CLL, MM, or AML                                                                                                                              |          | LV2020-001                           | TERMINATED                     | CLL, MM, AML                                | BIOLOGICAL: LAVA-051 BIOLOGICAL: interleukin 2                                                 |
| NCT04796441                    | Clinical study of universal CAR-γδ T cell injection in the treatment of patients with relapsed AML after transplantation                                                                            |          | CAR- <sup>131</sup> I T cell for AML | UNKNOWN                        | AML                                         | BIOLOGICAL: CAR-γδ T                                                                           |
| NCT04764513                    | Safety and efficiency of γδ T cell against hematological malignancies after allo-HSCT                                                                                                               |          | CHN-PLAGH-BT-062                     | RECRUITING                     | AML, ALL, MDS, lymphoma                     | BIOLOGICAL: ex vivo expanded γδ T-cell infusion                                                |
| NCT04735471                    | A phase 1 study of ADI-001 in B cell malignancies                                                                                                                                                   | GLEAN-1  | ADI-20200101                         | ACTIVE NOT RECRUITING          | FL, MCL, MZL, PMBCL, DLBCL, NHL             | BIOLOGICAL: ADI-001                                                                            |

| Cell type                                                    | Donor type | Cell modification?             | Sponsor                                             | Phases    | Funder type | Study type     | Start date |
|--------------------------------------------------------------|------------|--------------------------------|-----------------------------------------------------|-----------|-------------|----------------|------------|
| Vδ1-enriched γδ cells                                        | Allogeneic | No                             | Takeda                                              | PHASE 1/2 | INDUSTRY    | INTERVENTIONAL | 2023       |
| TCR αβ/CD19 depleted DLI with >10 <sup>6</sup> γδ T cells/kg | Allogeneic | No                             | Rigshospitalet, Denmark                             | PHASE 2/3 | OTHER       | INTERVENTIONAL | 2022       |
| γδ T cells conjugated to anti-CD20 antibody via DNA linkers  | Allogeneic | Yes, CD20 antibody conjugation | Acepodia Biotech, Inc.                              | PHASE 1   | INDUSTRY    | INTERVENTIONAL | 2023       |
| γδ T cells                                                   | Allogeneic | Yes, CD19-CAR                  | Chinese PLA General Hospital                        | PHASE 1/2 | OTHER       | INTERVENTIONAL | 2022       |
| γδ T cells                                                   | Unknown    | Yes, CAR-transduced            | Hebei Senlang Biotechnology Inc., Ltd.              | NA        | INDUSTRY    | INTERVENTIONAL | 2022       |
| Vγδ92 T cells                                                | Allogeneic | No                             | TC Biopharm                                         | PHASE 2   | INDUSTRY    | INTERVENTIONAL | 2022       |
| Vγ9δ2 T cells                                                | Allogeneic | No                             | TC Biopharm                                         | PHASE 1   | INDUSTRY    | INTERVENTIONAL | 2025       |
| γδ T cells                                                   | Allogeneic | Yes, NKG2DL-CAR                | CytoMed Therapeutics Pte Ltd.                       | PHASE 1   | INDUSTRY    | INTERVENTIONAL | 2024       |
| γδ T cells                                                   | Allogeneic | No                             | H. Lee Moffitt Cancer Center and Research Institute | PHASE 1   | OTHER       | INTERVENTIONAL | 2022       |
| Vδ1-enriched γδ cells                                        | Allogeneic | No                             | GammaDelta Therapeutics Limited                     | PHASE 1   | INDUSTRY    | INTERVENTIONAL | 2021       |
| γδ T cells                                                   | Unknown    | Yes, CAR-transduced            | Lava Therapeutics                                   | PHASE 1   | INDUSTRY    | INTERVENTIONAL | 2021       |
| γδ T cells                                                   | Unknown    | Yes, CAR-transduced            | Hebei Senlang Biotechnology Inc., Ltd.              | NA        | INDUSTRY    | INTERVENTIONAL | 2020       |
| γδ T cells                                                   | Allogeneic | No                             | Chinese PLA General Hospital                        | PHASE 1/2 | OTHER       | INTERVENTIONAL | 2021       |
| (Vδ1) γδ T cells                                             | Allogeneic | Yes, anti-CD20 CAR             | Adicet Therapeutics                                 | PHASE 1   | INDUSTRY    | INTERVENTIONAL | 2021       |

TABLE 2 (Continued)

| NCT number                                        | Study title                                                                                                                                | Acronym  | Other IDs                   | Study status                   | Conditions                                                         | Interventions                                                   |
|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------------------|--------------------------------|--------------------------------------------------------------------|-----------------------------------------------------------------|
| NCT04696705                                       | Allogeneic $\gamma\delta$ T cells immunotherapy in r/r non-Hodgkin's lymphoma (NHL) or peripheral T cell lymphomas (PTCL) patients         |          | IIT2020028-EC-3             | UNKNOWN                        | NHL, PTCL                                                          | BIOLOGICAL: ex vivo expanded allogeneic $\gamma\delta$ T cells  |
| NCT04008381                                       | Ex-vivo expanded $\gamma\delta$ T lymphocytes in patients with refractory/relapsed acute myeloid leukaemia                                 |          | WHUH-2009-V010              | UNKNOWN                        | AML                                                                | BIOLOGICAL: ex vivo expanded $\gamma\delta$ T lymphocytes       |
| NCT03939585                                       | Pre-emptive infusion of donor lymphocytes depleted of TCR + T cells + CD19+ B cells following ASCT                                         |          | CASE1Z19                    | WITHDRAWN                      | Allogeneic stem cell transplant candidate; AML, ALL, MDS, MPN, LPD | BIOLOGICAL: cellular therapy product                            |
| NCT03790072                                       | Ex-vivo expanded $\gamma\delta$ T-lymphocytes (OmnImmune <sup>®</sup> ) in patients with acute myeloid leukaemia (AML)                     |          | TCB-202-001 2018-000409-22  | COMPLETED                      | AML                                                                | BIOLOGICAL: OmnImmune <sup>®</sup>                              |
| NCT03533816                                       | Expanded/activated gamma delta T-cell infusion following hematopoietic stem cell transplantation and post-transplant cyclophosphamide      |          | IIT-2018-gamma-delta T cell | RECRUITING                     | AML, CML, ALL, MDS                                                 | BIOLOGICAL: EAGD T-cell infusion                                |
| NCT02656147                                       | Immunotherapy with CD19 CAR $\gamma\delta$ T-cells for B-cell lymphoma, ALL and CLL                                                        |          | Doing-004                   | UNKNOWN                        | Leukemia, lymphoma                                                 | BIOLOGICAL: anti-CD19-CAR $\gamma\delta$ T                      |
| NCT05237206                                       | Study of SUPLEXA in patients with metastatic solid tumours and haematologic malignancies                                                   |          | SUPLEXA-101                 | COMPLETED                      | Oncology                                                           | BIOLOGICAL: SUPLEXA                                             |
| NCT04702841                                       | CAR- $\gamma\delta$ T cells in the treatment of relapsed and refractory CD7 positive T cell-derived malignant tumors                       |          | 2019-ky012                  | UNKNOWN                        | CD7-positive T-cell malignancies                                   | DRUG: chimeric antigen receptor modified $\gamma\delta$ T cells |
| NCT04688853                                       | A study to investigate the safety and efficacy of TEG002 in relapsed/refractory multiple myeloma patients                                  |          | TEG002_MM_US_01             | UNKNOWN                        | MM                                                                 | BIOLOGICAL: TEG002                                              |
| NCT04028440                                       | $\gamma\delta$ T cells immunotherapy in patients with relapsed or refractory non-Hodgkin's lymphoma (NHL)                                  |          | QT2019001-EC-2              | UNKNOWN                        | NHL, CLL, PTCL                                                     | BIOLOGICAL: autologous $\gamma\delta$ T cells                   |
| <i>Antibody-based therapies</i>                   |                                                                                                                                            |          |                             |                                |                                                                    |                                                                 |
| NCT04887259                                       | Trial of LAVA-051 in patients with relapsed/refractory CLL, MM, or AML                                                                     |          | LV2020-001                  | TERMINATED (BUSINESS DECISION) | CLL, MM, AML                                                       | BIOLOGICAL: LAVA-051 BIOLOGICAL: interleukin 2                  |
| NCT04243499                                       | First-in-human study of ICT01 in patients with advanced cancer                                                                             | EVICION  | ICT01-101                   | RECRUITING                     | Solid tumor, adult hematopoietic/lymphoid cancer                   | BIOLOGICAL: IV ICT01 (BTN3A antibody)                           |
| <i>In vivo stimulation of gamma delta T cells</i> |                                                                                                                                            |          |                             |                                |                                                                    |                                                                 |
| NCT03862833                                       | Zoledronic acid in combination with interleukin-2 to expand V $\gamma$ 9 $\delta$ 2 T cells after T-replete haplo-identical allotransplant | HILDEGAZ | RC18_0419                   | COMPLETED                      | Hematopoietic stem cell transplantation                            | DRUG: IL2 DRUG: zoledronic acid                                 |
| <i>Observational studies</i>                      |                                                                                                                                            |          |                             |                                |                                                                    |                                                                 |
| NCT03885076                                       | Gamma delta T cells in AML                                                                                                                 |          | CCR4877                     | UNKNOWN                        | AML                                                                | PROCEDURE: blood collection and bone marrow aspirate            |
| NCT04911478                                       | Long-term follow-up study of allogeneic gamma delta ( $\gamma\delta$ ) CAR T cells                                                         |          | ADI-20200102                | ENROLLING BY INVITATION        | FL, MCL, MZL, PMBCL, DLBCL, NHL                                    | BIOLOGICAL: ADI-001                                             |

Note: The list was obtained from [clinicaltrials.gov](https://clinicaltrials.gov) on January 31, 2025.

Abbreviations:  $\gamma\delta$ , gamma delta; ALL, acute lymphoblastic leukemia; AML, acute myeloid leukemia; CAR, chimeric antigen receptor; CLL, chronic lymphocytic leukemia; CML, chronic myeloid leukemia; DLBCL, diffuse large B-cell lymphoma; FL, follicular lymphoma; LPD, lymphoproliferative disorder; MCL, mantle cell lymphoma; MDS, myelodysplastic syndrome; MM, multiple myeloma; MPN, myeloproliferative neoplasm; MZL, marginal zone lymphoma; NCT, national clinical trial number; N/A, not applicable; NHL, non-Hodgkin lymphoma; NK, natural killer; PMBCL, primary mediastinal B-cell lymphoma; PTCL, peripheral T-cell lymphoma; TCR, T-cell receptor.



| Cell type                                    | Donor type | Cell modification?    | Sponsor                                                  | Phases        | Funder type | Study type     | Start date |
|----------------------------------------------|------------|-----------------------|----------------------------------------------------------|---------------|-------------|----------------|------------|
| γδ T cells                                   | Allogeneic | No                    | Institute of Hematology & Blood Diseases Hospital, China | EARLY PHASE 1 | OTHER       | INTERVENTIONAL | 2020       |
| γδ T cells                                   | Allogeneic | No                    | Wuhan Union Hospital, China                              | PHASE 1       | OTHER       | INTERVENTIONAL | 2019       |
| γδ T- and NK-cell-enriched (TCR αβ-depleted) | Allogeneic | No                    | Leland Metheny                                           | PHASE 1       | OTHER       | INTERVENTIONAL | 2024       |
| γδ T cells                                   | Allogeneic | No                    | TC Biopharm                                              | PHASE 1       | INDUSTRY    | INTERVENTIONAL | 2018       |
| γδ T cells                                   | Allogeneic | No                    | University of Kansas Medical Center                      | PHASE 1       | OTHER       | INTERVENTIONAL | 2020       |
| γδ T cells                                   | Allogeneic | Yes; CD19 CAR         | Beijing Doing Biomedical Co., Ltd.                       | PHASE 1       | INDUSTRY    | INTERVENTIONAL | 2017       |
| NK, NK-T, and T cells                        | Autologous | Unknown               | Alloplex Biotherapeutics Inc                             | PHASE 1       | INDUSTRY    | INTERVENTIONAL | 2022       |
| γδ T cells                                   | Autologous | Yes, CD7 CAR          | PersonGen BioTherapeutics (Suzhou) Co., Ltd.             | EARLY PHASE 1 | INDUSTRY    | INTERVENTIONAL | 2020       |
| T cells                                      | Autologous | Yes, TCRγδ-transduced | Gadeta B.V.                                              | PHASE 1       | INDUSTRY    | INTERVENTIONAL | 2021       |
| γδ T cells                                   | Autologous | No                    | Institute of Hematology & Blood Diseases Hospital, China | EARLY PHASE 1 | OTHER       | INTERVENTIONAL | 2019       |
| N/A, CD1d/Vδ2 TCR bispecific antibody        | N/A        | N/A                   | Lava Therapeutics                                        | PHASE 1       | INDUSTRY    | INTERVENTIONAL | 2021       |
| N/A antibody-based                           | N/A        | N/A                   | ImCheck Therapeutics                                     | PHASE 1/2     | INDUSTRY    | INTERVENTIONAL | 2020       |
| N/A                                          | N/A        | N/A                   | Nantes University Hospital                               | PHASE 1       | OTHER       | INTERVENTIONAL | 2019       |
| N/A                                          | N/A        | N/A                   | Royal Marsden NHS Foundation Trust                       | N/A           | OTHER       | OBSERVATIONAL  | 2018       |
| γδ T cells                                   | Allogeneic | Yes, anti-CD20 CAR    | Adicet Therapeutics                                      | N/A           | INDUSTRY    | OBSERVATIONAL  | 2022       |

**TABLE 3** Comparison of immune effector cells for adoptive cell therapy.

|                                                                                | γδ T cells                                                                                                                                                         | αβ T cells                                                                                                                                                           | NK cells                                                                                                                      | INKT cells                                                                                                  |
|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Anticancer activity of unmanipulated cells                                     | Yes; MHC-presentation not required                                                                                                                                 | Yes; requires antigen presentation in the context of self-MHC molecules                                                                                              | Yes; recognizes stress signals via innate NKRrs                                                                               | Yes; CD1d-restricted. Classical MHC presentation not required                                               |
| Frequency in PB                                                                | Uncommon                                                                                                                                                           | Abundant                                                                                                                                                             | Less frequent                                                                                                                 | Rare                                                                                                        |
| Ease of manufacture                                                            | +                                                                                                                                                                  | +++                                                                                                                                                                  | +                                                                                                                             | +                                                                                                           |
| Feeder/antigen-presenting cell requirement for expansion                       | Not essential                                                                                                                                                      | May require                                                                                                                                                          | Often requires                                                                                                                | Often requires                                                                                              |
| Alloreactivity                                                                 | No                                                                                                                                                                 | Yes                                                                                                                                                                  | No                                                                                                                            | No                                                                                                          |
| Cell manipulation strategies in clinical trials                                | CAR, TCR transfer (TCRt)                                                                                                                                           | CAR T                                                                                                                                                                | CAR-NK                                                                                                                        | CAR-INKT                                                                                                    |
| "Off-the-shelf" potential                                                      | Yes                                                                                                                                                                | No; unless TCR deleted                                                                                                                                               | Yes                                                                                                                           | Yes                                                                                                         |
| In vivo persistence                                                            | Prolonged <sup>a</sup>                                                                                                                                             | Variable                                                                                                                                                             | Limited                                                                                                                       | Prolonged <sup>a</sup>                                                                                      |
| Risk of cytokine release syndrome (CRS)                                        | Low                                                                                                                                                                | Moderate/high                                                                                                                                                        | Low                                                                                                                           | Low                                                                                                         |
| Hematological conditions for CAR-based therapies (clinical trials or licensed) | B-cell malignancies<br>Multiple myeloma<br>T-cell malignancies<br>AML                                                                                              | B-cell lymphoma/leukemia<br>Multiple myeloma                                                                                                                         | Multiple myeloma<br>B-cell malignancies                                                                                       | B-cell lymphoma/leukemia                                                                                    |
| Currently approved cell therapy products <sup>b</sup>                          | —                                                                                                                                                                  | 7                                                                                                                                                                    | —                                                                                                                             | —                                                                                                           |
| Limitations                                                                    | <ul style="list-style-type: none"><li>Limited knowledge of most γδ TCR ligands.</li><li>Variable donor expansions.</li><li>Possible regulatory function.</li></ul> | <ul style="list-style-type: none"><li>Mostly limited to autologous use.</li><li>Potential for T cell exhaustion.</li><li>Limited activity in solid tumors.</li></ul> | <ul style="list-style-type: none"><li>Limited persistence in vivo models.</li><li>Lacks TCR-based clonal expansion.</li></ul> | <ul style="list-style-type: none"><li>Rare immune cells.</li><li>Potentially difficult to expand.</li></ul> |

Abbreviations: γδ, gamma delta; AML, acute myeloid leukemia; CAR, chimeric antigen receptor; MHC, Major Histocompatibility Complex; NK, natural killer; NKR, natural killer receptor; PB, peripheral blood; TCR, T-cell receptor.

<sup>a</sup>Based on limited clinical data.

<sup>b</sup>As of May 2025.

a subset of  $\gamma\delta$  T cells (mainly V $\delta$ 1 and V $\delta$ 3) producing amphiregulin (AREG) was identified in colorectal cancer samples in association with a “wound healing,” pro-tumor phenotype.<sup>79</sup> Notably, the study also found (in contrast to previous studies) that neither V $\delta$ 1 nor V $\delta$ 2 cells produced IL-17; rather V $\delta$ 3 T cells were the source of IL-17 in their model. Another recently published study also found no IL17A-expressing V $\delta$ 1 T cells on transcriptomic analyses of skin-resident  $\gamma\delta$  T cells.<sup>36</sup> From these experimental observations,  $\gamma\delta$ T17 induction and the pro-inflammatory  $\gamma\delta$  T-cell phenotype appear to be context and tissue-dependent, likely driven by conditions within the TME. These findings highlight the need for detailed phenotypic characterization of  $\gamma\delta$  T-cell products for clinical development.

The main obstacle to the development of  $\gamma\delta$  T-cell immunotherapy has been the difficulty in expanding this relatively rare T-cell subset to clinically useful cell numbers. The earliest methods employed T-cell mitogens such as PHA and Con A, which made clinical translation difficult. The V $\gamma$ 9V $\delta$ 2 subset is usually expanded from PBMCs with ZOL and stimulatory cytokines, but there are often significant variations in expansion efficiency between donors. While IL-2 and IL-15 have typically been used in expanding  $\gamma\delta$  T cells, novel combinations of cytokines have been investigated. A recent study demonstrated increased V $\gamma$ 9V $\delta$ 2 T-cell yield and enhanced effector functions against AML when  $\gamma\delta$  T cells were expanded in culture media containing transforming growth factor- $\beta$  (TGF- $\beta$ ).<sup>80</sup> Alternative methods include the use of agonist  $\gamma\delta$ TCR antibodies; however, an argument against this method is that chronic stimulation via the TCR during the expansion process may lead to functional exhaustion. Both artificial APCs and feeder cell-free methods have been evaluated in the expansion of  $\gamma\delta$  T cells<sup>81</sup> and new clinically relevant, Good Manufacturing Practice (GMP)-grade methods being developed for the expansion of the rarer V $\delta$ 1 subset from PBMCs include the depletion of  $\alpha\beta$  T cells followed by expansion with anti-CD3 (OKT3) and IL-15.<sup>50</sup> The Delta one T (DOT) cell protocol which uses a combination of cytokines (IL-4, IFN $\gamma$ , IL-21, and IL-1 $\beta$ ) and antiCD3 mAb (OKT3) expanded V $\delta$ 1 T cells which were cytotoxic against cancer cell lines.<sup>49</sup> An allogeneic cell therapy product (GDX012), based on the DOT protocol, is currently in clinical trials in patients with AML.<sup>82</sup> A 2023 paper described a protocol using IL-15, IL-18, anti-CD3, and anti-CD2 promoted the expansion of V $\delta$ 1 T cell with a strong antitumor phenotype.<sup>79</sup> With the recent interests in  $\gamma\delta$  T-cell-based therapeutics, it is crucial to advance research into the fundamental biology of these cells and the multifaceted functions of  $\gamma\delta$  T-cell subsets particularly their interaction with the tumor microenvironment. Further research priorities should be directed at understanding the basis for the significant variation in expansion efficiency between donors with a view to determining predictive factors for improved  $\gamma\delta$  T-cell expansion and quality.

T-cell exhaustion in conventional  $\alpha\beta$  T cells is well characterized, but emerging evidence suggests that canonical exhaustion markers such as PD-1 do not define exhaustion in  $\gamma\delta$  T cells as functional V $\delta$ 1 cells often express high levels of PD-1. Therefore, basic research into the characterization of  $\gamma\delta$  T-cell exhaustion signatures at the transcriptomic level is crucial to the identification of  $\gamma\delta$  T-cell-specific exhaustion markers, which will ultimately lead to the development of optimal cell products for the clinic.

## CONCLUSIONS

### $\gamma\delta$ T-cell therapy in hematology: Current gaps and potentials

Current CAR T-cell strategies involve manipulation of autologous T cells, and the manufacturing process often takes several weeks.

This is unsuccessful in up to 25% of patients due to poor T-cell fitness, effect of prior therapies, and suboptimal constitution of PBMCs.<sup>83</sup> In patients with rapidly progressing disease and those with T-cell dysfunction as a result of prior lymphotoxic treatments, autologous cell therapy strategies are undesirable; hence, the need for the development of allogeneic healthy donor-derived cell therapy products. The use of unconventional T cells ( $\gamma\delta$  T cells, iNKT, and mucosal-associated invariant T [MAIT] cells)<sup>84,85</sup> as well as innate immune cells such as NK cells as CAR effectors provides an attractive strategy given their broad anticancer reactivity and lack of alloreactivity. These immune cells offer the potential for “off-the-shelf” cell therapy products, which not only minimize the time to treatment but also potentially guarantee a standardized cell product. A summary of some immune effector cells being developed for adoptive cell therapy is shown in Table 3.

The feasibility of the off-the-shelf approach for  $\gamma\delta$  T cells is evidenced by the fact that clinical trials involving allogeneic  $\gamma\delta$  T cells have not reported GVHD as a complication.<sup>63,86</sup> Specifically,  $\gamma\delta$  T cells possess characteristics that are ideal for adoptive cell therapy. Besides their lack of alloreactivity,  $\gamma\delta$  T-cell-based CAR therapy has been associated with a lower risk of CRS compared with conventional  $\alpha\beta$  T cells,<sup>87</sup> and data from a Phase I trial indicated prolonged persistence in vivo.<sup>88</sup> Furthermore, given the natural localization of certain  $\gamma\delta$  T-cell subsets to tissues, it is thought that  $\gamma\delta$  T cells would be effective in the solid tumor microenvironment, hence the recent increase in  $\gamma\delta$  T-cell trials against solid tumors. The tissue tropism of  $\gamma\delta$  T cells would also be crucial in certain hematological malignancies, for instance, in extramedullary myeloma, where optimal immune cell infiltration is required for tumor eradication. Key limitations in the use of  $\gamma\delta$  T cells for adoptive cell therapy include donor variability in  $\gamma\delta$  T-cell subsets and expansion, limited knowledge of the natural ligands of the  $\gamma\delta$  TCR, and their diverse functions in disease pathogenesis.

In the post-allogeneic SCT setting, the administration of DLIs is currently an established therapy for mixed donor chimerism<sup>89</sup> but this is often associated with an increased risk of GVHD. The lack of MHC-restriction by  $\gamma\delta$  T cells potentially provides an opportunity to achieve GvL effects without the risk of GVHD in the post-allogeneic SCT setting. Therefore, the use of ex vivo expanded donor-derived  $\gamma\delta$  T cells post-allogeneic SCT would be a rational strategy to deepen clinical responses and as an effective therapy against post-transplant infections.

The hematology cell therapy space has seen tremendous progress with all seven autologous FDA-approved CAR T-cell products, being for B-cell malignancies and MM.<sup>90</sup> Despite these advancements, there are still significant unmet needs for advanced cell therapies against many hematological malignancies. For instance, there are currently limited treatment options for relapsed/refractory AML, and the lack of an ideal AML-associated antigen limits the development of safe CAR-based therapy. Given the ample preclinical evidence of unmanipulated  $\gamma\delta$  T-cell activity against AML,<sup>91</sup> the use of ex vivo allogeneic  $\gamma\delta$  T cells could be an effective therapeutic strategy for AML as well as other malignancies where no ideal tumor target exists. The anticancer activity of unmanipulated  $\gamma\delta$  T cells in this setting is underpinned by the recognition of neoantigens and “stress signals” by the myriad of innate receptors and the  $\gamma\delta$ TCR. However, tumor specificity and anticancer efficacy are enhanced by CAR-based approaches, and this is preferable where an ideal target antigen exists. To improve efficacy and ensure  $\gamma\delta$  T-cell persistence in vivo, next-generation  $\gamma\delta$  CAR engineering approaches will require a combination of strategies including multi-antigen targeting, novel CAR-binding domains, and cytokine armoring. Given the pivotal role of IL-15 and other stimulatory cytokines in the expansion and maintenance of  $\gamma\delta$  T cells, the development of cytokine-armoured

CARs is expected to improve persistence in vivo. However, concerns around cytokine-induced uncontrolled T-cell proliferation and other toxicities need to be addressed with novel strategies including inducible cytokine expression or the incorporation of a suicide gene in the CAR construct.

## Future perspectives

The recent explosion of scientific interest in  $\gamma\delta$  T cells reflects the realization of their crucial role as a first line of defense against certain microbes and transformed cells. Despite the increased  $\gamma\delta$  T-cell research focus, many fundamental questions remain unanswered. These include an understanding of the factors underlying specific tissue localization of  $\gamma\delta$  T-cell subsets, identification of  $\gamma\delta$  TCR ligands, and the complex roles of  $\gamma\delta$  T cells in health and disease pathogenesis. The considerable experimental evidence for  $\gamma\delta$  T-cell antitumor activity against a broad range of solid and hematological cancers makes  $\gamma\delta$  T cells ideal candidates for translation into cell therapy products for use in the clinic. However, there is a need for optimization of  $\gamma\delta$  T-cell expansion methods with a view to ensuring adequate T-cell fitness and avoidance of undesirable immune cell populations. As knowledge of the biology of  $\gamma\delta$  T-cell subsets advances, it is likely that new expansion methods will be developed that optimize desirable effector functions.

While unmanipulated  $\gamma\delta$  T cells demonstrate potent effector functions against tumors in experimental models, further enhancement and redirection of function are achieved by transduction of CARs.<sup>92,93</sup> Feasibility for this approach in the clinic has been demonstrated in prostate, triple negative breast, and hepatocellular preclinical models.<sup>94,95</sup> So far, early results from a clinical trial of allogeneic CD20 CAR  $\gamma\delta$  T cells have demonstrated safety and tolerability in a heavily pretreated group of patients with B-cell malignancies.<sup>86</sup> Another cell manipulation strategy currently investigated in a clinical trial is  $\gamma\delta$ TCR transfer (NCT04688853), which involves transduction of a specific  $\gamma\delta$  TCR onto autologous  $\alpha\beta$  T cells. In addition to current established CAR redirection strategies, research efforts are being channeled into strategies that optimize persistence and effector functions. These include the arming of CAR T cells with cytokines<sup>80,96,97</sup> and the conditional activation of cytokine signaling following CAR engagement and granzyme release.<sup>98</sup> It is expected that the  $\gamma\delta$  T-cell therapy field will see an expansion of advanced cell manipulation strategies in the future. Cell-based immunotherapy against solid tumors is currently limited by several unique obstacles including impaired immune cell infiltration due to the immunosuppressive tumor microenvironment. As  $\gamma\delta$  T-cell subsets have tissue distributions likely indicating adaptations to specific tissues, it is expected that  $\gamma\delta$  T cells would be ideal immune effectors against solid tumors either alone or in combination with checkpoint inhibitors. Evidently, well-designed clinical trials are required to determine the efficacy of  $\gamma\delta$  T cells in solid cancers, and the results of ongoing trials are awaited.

In summary,  $\gamma\delta$  T cells are versatile immune cells with immense potential for development as cell therapy products against a broad range of cancers; however, further fundamental research is required to fully elucidate the complex roles of  $\gamma\delta$  T cells in cancer. A combination of detailed phenotypic characterization of  $\gamma\delta$  T cells, optimized expansion methods, and cell manipulation strategies would likely lead to the development of highly effective  $\gamma\delta$  T-cell therapy products. While most recent advancements in cell therapy, particularly CAR T-cell therapy, have been hematology-led, it is anticipated that  $\gamma\delta$  T cells would also open new avenues for the development of adoptive cell therapies for solid tumors in the near future.

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## AUTHOR CONTRIBUTIONS

**Charles Agbuduwe:** Conceptualization; methodology; writing—original draft; writing—review and editing; data curation; resources. **John Maher:** Writing—review and editing; supervision; methodology. **John Anderson:** Writing—review and editing; supervision; methodology.

## CONFLICT OF INTEREST STATEMENT

C.A.: no conflicts. J.M.: chief scientific officer, scientific founder, and shareholder of Leucid Bio. J.A.: founder stock in Autolus Ltd. and holds patents related to CAR-T technology.

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